

GLYCOLYSIS PIPELINE

G6P → F6P → G6P → F6P → G6P → H2P → MGP → G6P

PFK DEFICIENCY
= TARUI'S DISEASE
= GSD TYPE VII

TPI DEFICIENCY
- SEVERE CNS + NEUROMUSCULAR
- MOST SEVERE non-erythroid involvement of all enzymopathies

PGK1 - X-LINKED
Only X-linked glycolytic deficiency (besides G6PD)

P5N DEFICIENCY
- inhibits P5N
→ SAME STIPPLING

LEAD POISONING
- inhibits P5N
→ SAME STIPPLING

L-6 BLN DEFICIENCY
RNA fragments accumulated

SPLEEN CAGE
BETOR

RETICULOCYTES ↑ AFTER SPLENECTOMY
↑ Hgb - modest
rising reticulocytes - paradox

THINK ENZYMOPATHY

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COOMBS-NEGATIVE DIAGNOSTIC SIGN

IRON OVERLOAD
- even without transfusion
→ chelation needed

IRON ATP OUTPUT VALVE

FOREMAN
DIAGNOSIS DELAYED

FIRST DIAGNOSIS: YOUNG PREGNANT WOMAN

RED CELL MACHINE

MITAPIVAT (PYRUKYND)
MISSENSE ONLY - at least one

PYRUVATE KINASE (PK) - LAST STEP

O₂ DELIVERY ENHANCED

↑ 2,3-BPG

↑ O₂

STMIH-TORSCDEYS
inhibits P5N N-@snehsux

1. Glycolysis pipeline

WHAT YOU SEE

Pipeline of G6P-F6P-...-pyruvate

WHAT IT ENCODES

Mature RBCs lack mitochondria, so glycolysis (Embden-Meyerhof) is their ONLY ATP source. Enzyme defects along this pipeline cause chronic nonspherocytic hemolytic anemia; severity depends on which step and residual activity.

BOARD TRAP

RBC enzymopathies cause Coombs-NEGATIVE, nonspherocytic hemolysis — distinguishing them from autoimmune (Coombs+) and membrane disorders (spherocytes).

2. Pyruvate kinase (PK) deficiency — last step

WHAT YOU SEE

PK valve at pipeline end

WHAT IT ENCODES

PK deficiency is the MOST COMMON glycolytic enzymopathy and the second most common enzymatic cause of hemolysis after G6PD. Autosomal recessive; ATP depletion causes chronic nonspherocytic hemolytic anemia, often from birth.

BOARD TRAP

PK deficiency is autosomal RECESSIVE and Coombs-negative — contrast with G6PD (X-linked, episodic, bite cells/Heinz bodies).

3. 2,3-BPG accumulation — right-shift

WHAT YOU SEE

Side tank: increased 2,3-BPG

WHAT IT ENCODES

In PK deficiency, the upstream block raises 2,3-BPG, which right-shifts the O₂-hemoglobin curve and ENHANCES O₂ delivery to tissues. Patients tolerate anemia better than the Hb level predicts.

BOARD TRAP

PK patients are often less symptomatic than their Hb suggests because high 2,3-BPG improves tissue oxygen unloading.

4. Reticulocytosis after splenectomy — paradox

WHAT YOU SEE

Spleen cage releasing reticulocytes

WHAT IT ENCODES

PK deficiency relies partly on residual mitochondrial ATP in reticulocytes; the spleen culls them. After SPLENECTOMY the reticulocyte count paradoxically RISES (more survive) even as hemolysis improves.

BOARD TRAP

A rising reticulocyte count after splenectomy in PK deficiency is expected and not treatment failure — a classic paradox.

5. Mitapivat (Pyrukynd)

WHAT YOU SEE

Mitapivat tablet, smiling RBC

WHAT IT ENCODES

Mitapivat is an oral PK ACTIVATOR that increases enzyme activity, raises hemoglobin, and reduces hemolysis/transfusion need in PK deficiency. It is the targeted therapy for missense (residual-activity) genotypes.

BOARD TRAP

Mitapivat works only when there's some residual enzyme to activate (missense alleles) — null/null genotypes respond poorly.

6. PFK deficiency — Tarui disease (GSD VII)

WHAT YOU SEE

Muscle figure, dark urine

WHAT IT ENCODES

Phosphofructokinase deficiency (Tarui, glycogen storage disease type VII) combines a myopathy with exertional myoglobinuria AND a compensated hemolytic anemia, because PFK is shared by muscle and RBC glycolysis.

BOARD TRAP

Exercise intolerance + myoglobinuria + hemolysis points to PFK (Tarui), distinct from McArdle (GSD V, no hemolysis).

7. TPI deficiency — severe CNS + neuromuscular

WHAT YOU SEE

CNS/brain figure

WHAT IT ENCODES

Triosephosphate isomerase deficiency is the MOST SEVERE glycolytic enzymopathy: hemolytic anemia PLUS progressive neuromuscular and CNS degeneration, usually fatal in early childhood.

BOARD TRAP

TPI deficiency is unique among enzymopathies for severe neurologic involvement — not just hematologic.

8. PGK1 deficiency — X-linked

WHAT YOU SEE

X-chromosome figure

WHAT IT ENCODES

Phosphoglycerate kinase deficiency is the ONLY X-linked glycolytic enzymopathy (besides G6PD), causing hemolysis with variable myopathy and CNS dysfunction.

BOARD TRAP

Among glycolytic enzymes PGK is the X-linked one; most others (PK, TPI, PFK) are autosomal recessive.

9. Basophilic stippling — lead poisoning

WHAT YOU SEE

RBCs with blue stippling

WHAT IT ENCODES

Lead inhibits pyrimidine-5'-nucleotidase (and ferrochelatase/ALAD), leaving ribosomal RNA fragments that appear as coarse basophilic stippling. Causes hemolysis plus neurologic and GI symptoms.

BOARD TRAP

Coarse basophilic stippling = lead poisoning (P5'N inhibition); fine stippling is nonspecific. Don't confuse with reticulocyte polychromasia.

10. P5'N deficiency — same blood picture as lead

WHAT YOU SEE

P5N deficiency figure

WHAT IT ENCODES

Inherited pyrimidine-5'-nucleotidase deficiency mimics lead poisoning: RNA fragments accumulate causing basophilic stippling and hemolysis, but WITHOUT elevated lead.

BOARD TRAP

Basophilic stippling with normal lead level = hereditary P5'N deficiency; both share the mechanism (failure to clear RNA).

11. Iron overload without transfusion

WHAT YOU SEE

Vault of iron

WHAT IT ENCODES

Chronic hemolytic enzymopathies (esp. PK deficiency) cause iron overload from increased absorption even WITHOUT transfusions; chelation may be needed.

BOARD TRAP

Don't assume iron overload in hemolysis requires transfusions — ineffective erythropoiesis drives absorption independently.

12. Coombs-negative — diagnostic sign

WHAT YOU SEE

Banner: Coombs-negative

WHAT IT ENCODES

Enzymopathies and membrane defects are Coombs (direct antiglobulin) NEGATIVE, separating them from autoimmune hemolytic anemia. A negative DAT in chronic hemolysis redirects work-up to enzyme assays.

BOARD TRAP

A negative Coombs in chronic hemolysis rules OUT autoimmune cause and points to enzyme/membrane/hemoglobin disorders.

13. Pressure accumulation side tank

WHAT YOU SEE

Side tank on pipeline

WHAT IT ENCODES

Upstream blocks in glycolysis cause buildup of intermediates (e.g., 2,3-BPG in PK deficiency) which both modulate O2 affinity and indicate the level of the enzymatic lesion.

BOARD TRAP

The pattern of accumulated intermediates localizes the defective enzyme — useful when interpreting metabolic studies.

14. Young pregnant woman — first presentation

WHAT YOU SEE

Pregnant figure, diagnosis delayed

WHAT IT ENCODES

Mild PK deficiency can stay subclinical for years and first declare itself when hemolysis is unmasked by pregnancy or another stressor, with delayed diagnosis.

BOARD TRAP

A chronic well-compensated enzymopathy may present late under physiologic stress — absence of childhood anemia doesn't exclude it.